Т11 $H_0: \xi \sim p_0(x) = 1 \{(0,1)\}$

$H_1: \xi \sim p_1(x) = \frac{e^{1-x}}{e-1} \{(0,1)\}$

а) $n = 1$, ур. значимости α α_1, α_2, W – ?

$$l(x) = \frac{L_1}{L_0} = \frac{p_1(x)}{p_0(x)} = \frac{e^{1-x}}{e-1} \geq C, \quad e^{-x} \geq B \Rightarrow$$

$\Rightarrow G: x \leq A$

$$P(x \leq A | H_0) = \alpha$$

$$\int_0^A 1\,dx = \alpha; \quad A = \alpha; \quad G: x \leq \alpha$$

$$W = P(x \leq A | H_1) = \int_0^A \frac{e^{1-x}}{e-1}\,dx = \frac{e}{e-1} \cdot \frac{e^{-x}}{(-1)}\Big|_0^A =$$

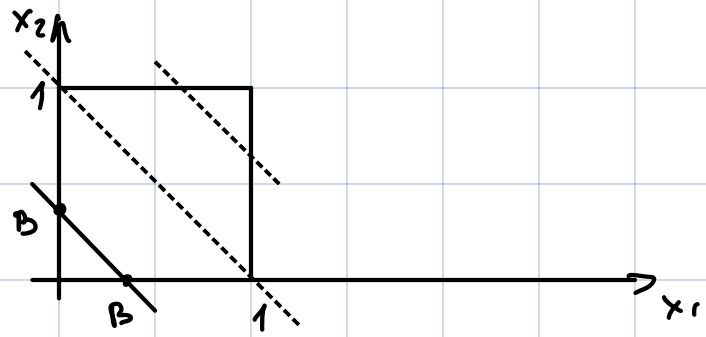
$$= \frac{e}{e-1}(1 - e^{-\alpha})$$

$$\boxed{\begin{aligned} \alpha_1 &= \alpha \\ \alpha_2 &= 1 - W \end{aligned}}$$

б) $n = 2$, α

$$l = \frac{L_1}{L_0} = \frac{\frac{e^{1-x_1}}{e-1} \cdot \frac{e^{1-x_2}}{e-1}}{1 \cdot 1} = \frac{e^{2-x_1-x_2}}{(e-1)^2} \geq C$$

$$G: \; x_1 + x_2 \le B$$



$$P(x_1 + x_2 \le B \mid H_0) = \alpha$$

$$\iint\limits_{\substack{x_1+x_2 \le B \\ 0 \le x_1 \le 1 \\ 0 \le x_2 \le 1}} 1 \cdot 1 \, dx_1 dx_2 = \alpha$$

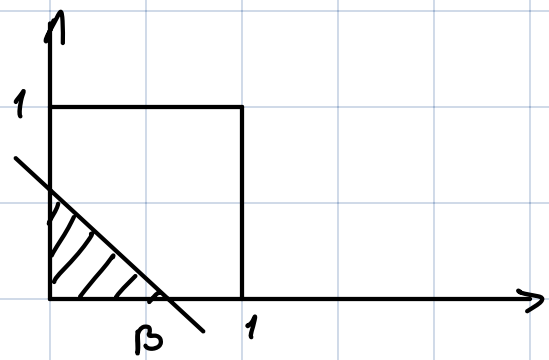
$$\frac{B^2}{2} = \alpha \; ; \; B = \sqrt{2\alpha} \quad (B \le 1)$$

$$\sqrt{2\alpha} \le 1 \; ; \; \boxed{\alpha \le \frac{1}{2}}$$

$$G: \; x_1 + x_2 \le B = \sqrt{2\alpha}$$

$$W = P(x_1 + x_2 \le B \mid H_1) = \iint\limits_{\substack{x_1+x_2 \le B \\ 0 \le x_i \le 1}} \frac{e^{1-x_1} \cdot e^{1-x_2}}{(e-1)^2} \, dx_1 dx_2 =$$

$$= \frac{e^2}{(e-1)^2} \int_0^B dx_1 \cdot \int_0^{B-x_1} e^{-x_1} \cdot e^{-x_2} \, dx_2 =$$



$$= \left(\frac{e}{e-1}\right)^2 \cdot \int_0^B e^{-x_1} \cdot \left(1 - e^{-B+x_1}\right) dx_1 =$$

$$= \left(\frac{e}{e-1}\right)^2 \left(1 - e^{-B} - e^{-B} \cdot B\right)$$

$$\boxed{\begin{aligned} \alpha_1 &= \alpha \\ \alpha_2 &= 1 - W \end{aligned}}$$

c) $$l = \frac{L_1}{L_0} = \prod_{i=1}^{n} \frac{p_1(x_i)}{p_0(x_i)} \ge C$$

$$\ln l = \sum_{i=1}^{n} \underbrace{\ln\left(\frac{p_1}{p_0}\right)}_{\eta_i} \ge \ln C$$

$$\sum_{i=1}^{n} \ln p_1(x_i) - \sum_{i=1}^{n} \ln p_0(x_i) = \sum_{i=1}^{n} \ln\left(\frac{e^{1-x}}{e-1}\right) - 0 \geq \ln c$$

$$\sum_{i=1}^{n} (1-x_i) - n\ln(e-1) \geq \ln c$$

$$G: \quad \sum_{i=1}^{n} x_i \leq A \qquad\qquad \bar{x} \leq B$$

$$\frac{\bar{x} - M\xi}{\sqrt{D\xi}}\sqrt{n} \rightsquigarrow N(0,1) \qquad (\text{ЦПТ})$$

$$H_0: \ M\xi = \frac{1}{2}, \quad D\xi = \frac{1}{12}$$

$$P(\bar{x} \leq B \mid H_0) = P\left(\frac{\bar{x} - 1/2}{\sqrt{1/12}}\sqrt{n} \leq \underbrace{\frac{B - 1/2}{\sqrt{1/12}}\sqrt{n}}_{u_\alpha}\right)$$

$$B = \frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}}$$

$$G: \ \bar{x} \leq \frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}}$$

$$W = P(\bar{x} \leq B \mid H_1)$$

$$H_1: \ M\xi = \int_0^1 x\, p_1(x)\,dx = \frac{e-2}{e-1}$$

$$M\xi^2 = \int_0^1 x^2 p_1(x)\,dx = \frac{2e-5}{e-1}$$

$$D\xi = \frac{e^2 - 3e + 1}{(e-1)^2}$$

$$P\left(\frac{\bar{x} - M\xi}{\sqrt{D\xi}}\sqrt{n} \le \frac{B - M\xi}{\sqrt{D\xi}}\sqrt{n}\right) = \int_{-\infty}^{\gamma} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

состоят. критерия

$$W \xrightarrow[n \to \infty]{} 1 \qquad \forall \text{ распр. из. } H_1$$

$$\gamma = \frac{\frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}} - 0{,}418}{\sqrt{D\xi}}\sqrt{n} = \frac{\frac{u_\alpha}{\sqrt{12}} + 0{,}082\sqrt{n}}{\sqrt{D\xi}} \xrightarrow[n \to \infty]{} \infty$$

Критерий состоятелен

d) $G: X_{min} < C \qquad H_0: \xi \sim R(0,1)$

$\alpha_2 \to \min$

$\alpha_1 \le \alpha$

$$P(X_{min} < C \mid H_0) = 1 - (1 - F(C)^n) \le \alpha$$

$$1 - (1 - C)^n \le \alpha \Rightarrow 1 - \sqrt[n]{1-\alpha} \ge C$$

$$\alpha_2 = P(X_{min} > C \mid H_1) = 1 - P(X_{min} \le C \mid H_1) = (1 - \Phi(C))^n =$$

$$= \left(1 - \frac{e}{e-1}(1 - e^{-C})\right)^n$$

$$(\alpha_2)'_c = n\left(1 - \frac{e}{e-1}\left(1 - e^{-c}\right)\right)^{n-1}\left(-\frac{e^{1-c}}{e-1}\right) = 0$$

$$\frac{e}{e-1}\left(1 - e^{-c}\right) = 1 \Rightarrow e^{-c} = e^{-1} \Rightarrow c = 1$$

$$\alpha_1 \le \alpha \Rightarrow 1 - \sqrt[n]{1-\alpha} \ge c$$

$$0 < 1 - \alpha < 1 \Rightarrow$$

$$\Rightarrow 0 < \sqrt[n]{1-\alpha} < 1$$

$$\Rightarrow 0 < 1 - \sqrt[n]{1-\alpha} < 1$$

Получаем $c > 1 - \sqrt[n]{1-x}$

$$\frac{d^2\alpha_2}{dc^2} = n(n-1)\left(1 - \frac{e}{e-1}\left(1 - e^{-c}\right)\right)^{n-2}\left(-\frac{e^{1-c}}{e-1}\right)^2 +$$

$$+ n\left(1 - \frac{e}{e-1}\left(1 - e^{-c}\right)\right)^{n-1}\left(\frac{e^{1-c}}{e-1}\right) > 0$$

$\Rightarrow$ c – минимум $\quad c \le 1 - \sqrt[n]{1-\alpha} \Rightarrow$

$$c = 1 - \sqrt[n]{1-\alpha}$$

$$\alpha_1 = \alpha$$

$\Rightarrow$ $G: \; x_{min} < 1 - \sqrt[n]{1-\alpha}$

$$W = P(X_{min} < c \mid H_1) = \left\{ H_1 : \{ \sim \frac{e}{e-1} e^{-x} \{(0,1)\} \right.$$

$$\left. F_1(x) = \frac{e}{e-1}(1-e^{-x}) \right\} =$$

$$= 1 - \left(1 - \frac{e}{e-1}(1-e^{-c})\right)^n = 1 - \left(1 - \frac{e}{e-1}\left(1 - e^{-1+\sqrt[n]{1-\alpha}}\right)\right)^n$$

$$\begin{cases} c = -1 + (1-\alpha)^{1/n} = \frac{1}{n}\ln(1-\alpha) + o\left(\frac{1}{n}\right) \\ e^{-c} = e^{\frac{1}{n}\ln(1-\alpha) + o\left(\frac{1}{n}\right)} = 1 + \frac{1}{n}\ln(1-\alpha) + o\left(\frac{1}{n}\right) \end{cases}$$

$$W = 1 - \left(1 + \frac{1}{n} \cdot \frac{e}{e-1}\ln(1-\alpha) + o\left(\frac{1}{n}\right)\right)^n \longrightarrow$$

$$\xrightarrow[n \to \infty]{} 1 - \exp\left(\frac{e\ln(1-\alpha)}{e-1}\right) = 1 - (1-\alpha)^{\frac{e}{e-1}} \neq 1$$

$\Rightarrow$ критерий не состоятелен